

CMPUT 175
Introduction to Foundations of Computing II

Fall 2026 LEC A1 MWF 9:00 – 9:50 AM NRE 1-001

Fall 2026 LEC A2 MWF 11:00 – 11:50 AM NRE 2-003

Slides written by Dr. Osmar Zaiane and revised by Dr. Euijin Choo

DR. EUIJIN CHOO

EUIJIN@UALBERTA.CA

Instructor Information

- Dr. Euijin Choo, Assistant Professor
- Main research area: Data-driven Security
- Personal web: alleychoo.github.io

- Office: University Commons, 7-181

- E-mail: euijin.choo@ualberta.ca
- Email policy: Use your UofA account and subject line should be "**[CMPUT175 Section#] Your purpose for email**"
  - E.g., [CMPUT175 A1] question about midterm1
  - If not, email response may be delayed or not answered

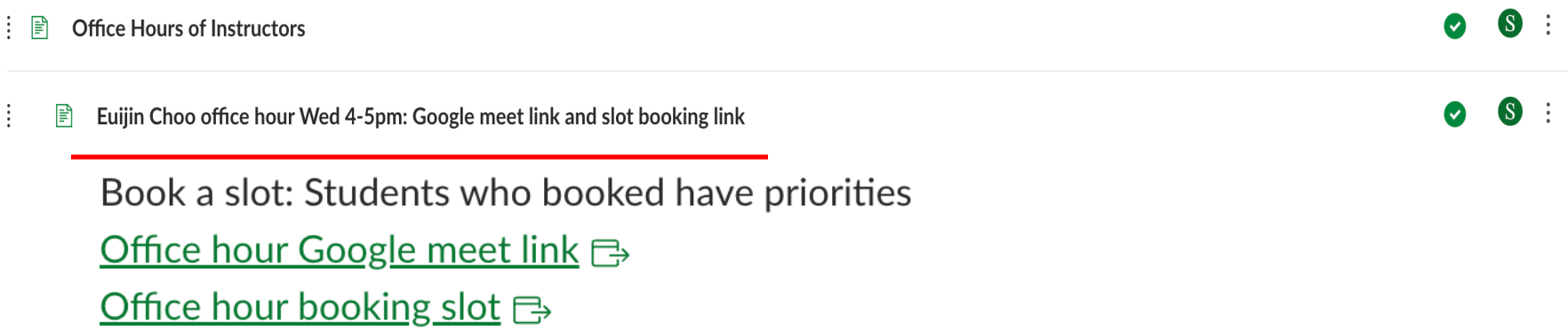
- **Edforum** may be more useful

Instructor Information

- Office hours: 4:00-5:00 pm, every Wed
- In-person
- Google Meet Link: meet.google.com/vvy-dgna-ubm
  - Feel free to join during the time, but book your slot(s) for a better arrangement:

    https://docs.google.com/spreadsheets/d/1liNWgwOYHUnYav2Oa349PhyflXxmu816klhG5vWlHq8/edit?usp=sharing

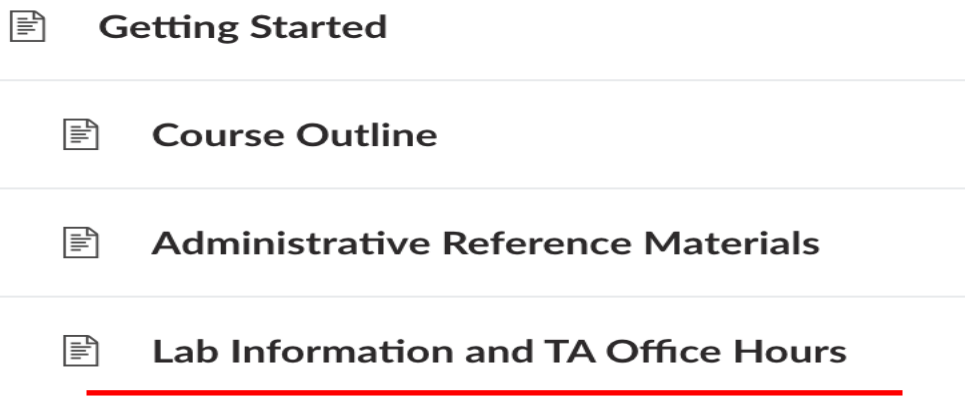
    - Students who booked the slots will have priorities



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Teaching Assistants

- 10 TAs
- All TAs have office hours (In-person)
- You can attend any of them



Course Structure

- 3 lecture hours + 3 lab hours per week + online videos for post quizzes for each week

Course Schedule

- Lectures

	8-9AM	9-10AM	10-11	11-12	12-1PM
Monday		A1		A2	
Tuesday					
Wednesday		A1		A2	
Thursday					
Friday		A1		A2	

A1	NRE 1-001
A2	NRE 2-003

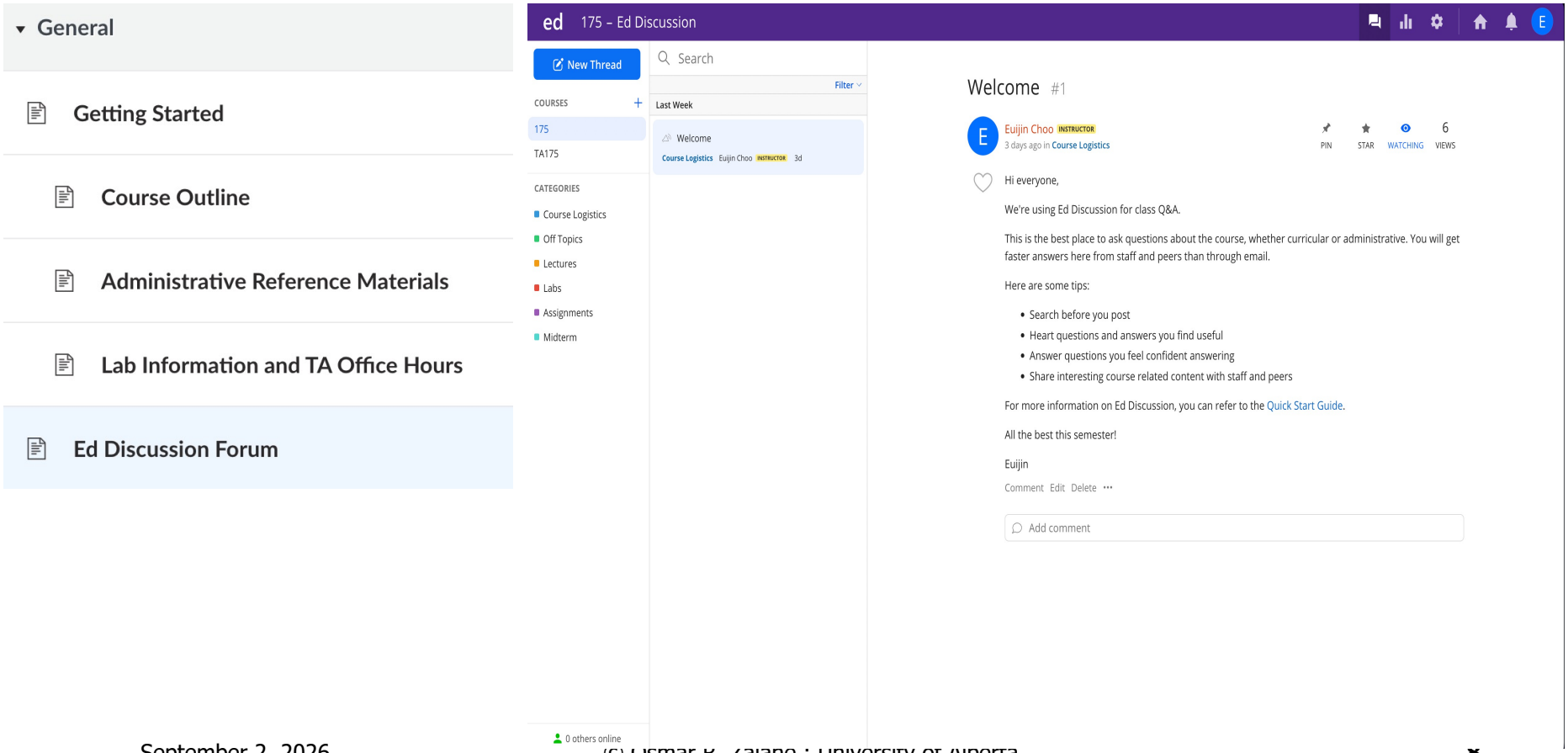
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Course Web Site

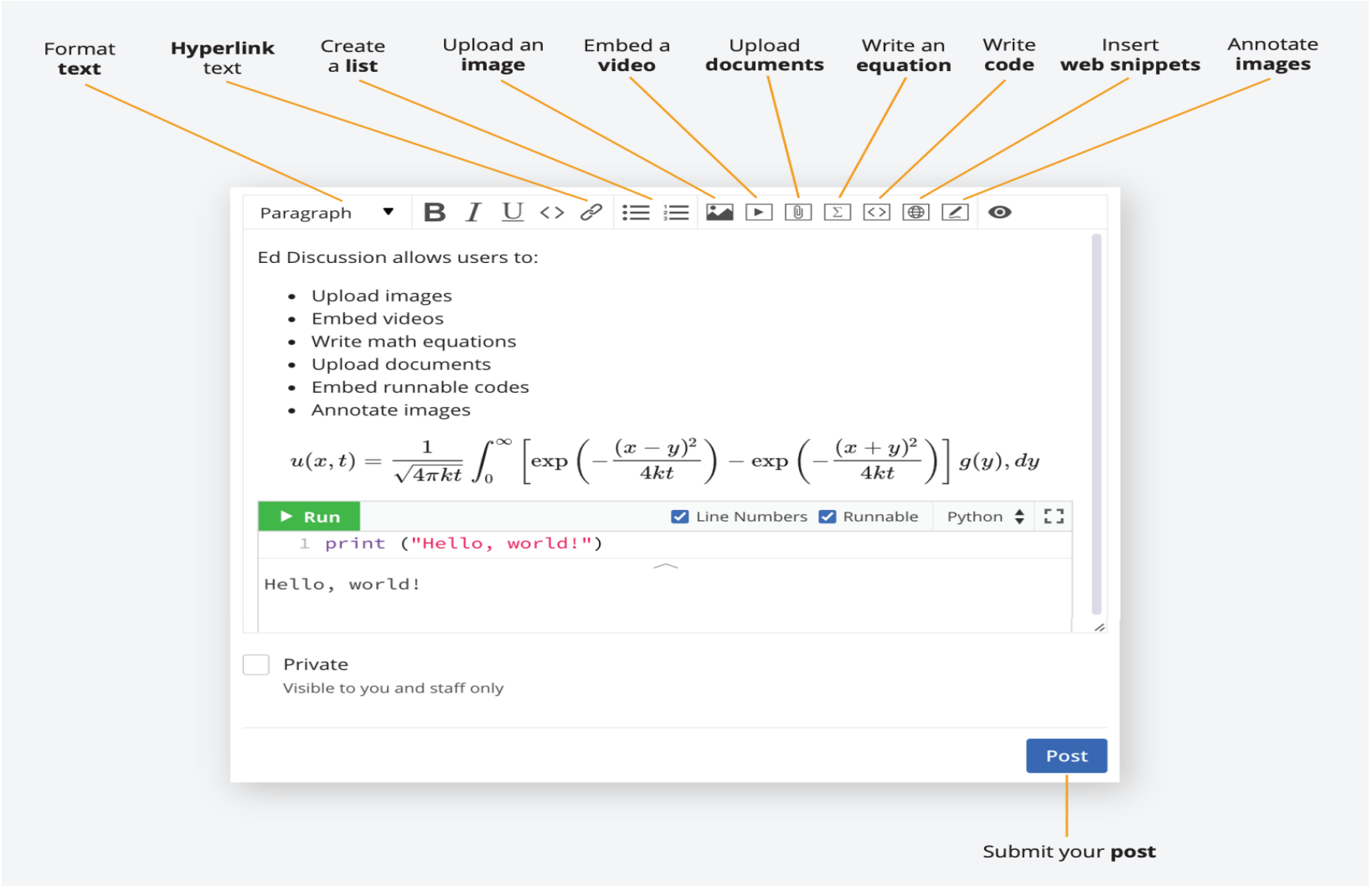
- Course Contents are on Canvas – (with CCID login)
- Frequently check the Canvas
- https://canvas.ualberta.ca/courses/34704
- Do not share course materials outside class without permission
- Slides, Videos, Post-quizzes, Assignments, resources, etc.
- Use the course forum (Ed Forum)
- Lab Submission Platform – (with participant ID login)

Course Discussion – Ed Forum

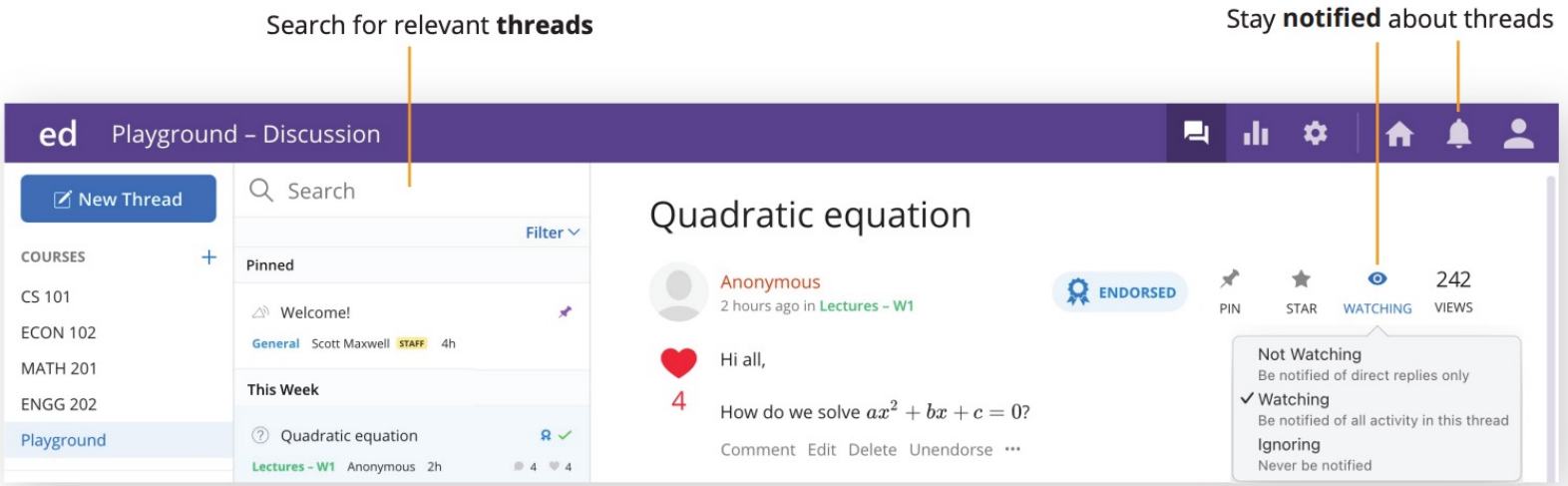
- https://edstem.org/us/courses/100706/discussion
- Ad-hoc Announcements may be communicated only via Edforum
- Join Ed Forum today!



Course Discussion – Ed Forum



Course Discussion – Ed Forum



General Schedule per week



Week 1
W: Syllabus + Intro
F: Python 101
Week 2
W: Python 101
F: Hangman
Week 3
M: Jupyter
W: Jupyter Notebook
F: Algorithm Analysis
Week 4
M:Algorithm Analysis
W: Algorithm Analysis
F: ADT
Week 5
M: Stack
F: Stack

Week 6
M: Stack
W: Exception
F: Exception
Week 7
W: Midterm 1 (Oct 14)
F: Queue
Week 8
M: Queue
W :Linked Lists
F: Linked Lists
Week 9
M: Doubly-Linked Lists
W: Doubly-Linked Lists
F: Recursion
Week 10
M: Search
W: Search
F: Midterm 2 (Nov 6)

Week 11: Reading Week
Week 12
W: Sorting
F: Sorting
M: Sorting
Week 13
M: Sorting
W: Trees
F: Trees
Week 14
M: Binary search trees
W: Binary search trees
F: Hash tables

Week 15
M: Midterm 3 (Dec 7)



Course Schedule

(Tentative, subject to changes)



Attendance is mandatory
There are weekly exercises to be done in the lab.

Midterm 1 **week 7 (Wed Oct 14th 2026)**

Midterm 2 **week 10 (Mon Nov 6th 2026)**

Midterm 3 **week 15 (Mon Dec 7th 2026)**

No class on Public holidays

No class from Nov 9th to 13th (Reading week)

Labs

- All guidelines and news are on the web
  - Ed forum/Canvas
  - Please read carefully to avoid unpleasant surprises
- Your TAs are your Instructors for the lab

Lab Assignments

- 3-weeks cycle
- Lab assignment will be released on Wed
- Lab assignment is due on Fri of the following week
  - Should be submitted to lab submission platform
- Do the demo during your lab time in the following week
- No demo – Zero score

Lab Submission Platform

- Instruction

- Need your participant IDs for using the platform (Read direction on Canvas)
  - Participant IDs appear in “comment” in Lab Login Information page

- No copy & paste
- Need to work in the platform

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Lab Submission Platform

- Unlimited Trials
- Feedbacks

Labs

3 hour labs per week

	14:00 - 16:50
	Lab
Tuesday	D1 (UCOMM3-130)
	17:00 - 19:50
Tuesday	D2 (UCOMM3-130)
Wednesday	D3 (UCOMM2-086)
Thursday	D4 (UCOMM2-086)

Labs

- Labs start next week (Sep 8th, 2026)
- First Lab
  - Lab Intro
  - General information: How TAs will grade etc.
  - Help session for the lab submission platform

- Other Labs
- Lab assignment demos (Sign up for the s every week)
- Assignment demos (Notified by TA)



General Schedule

September 2026

Rx: Lab Release Dx: Lab is Due Lx: Do Lab Demo

Monday	Tuesday	Wednesday	Thursday	Friday
	1	2 Syllabus R1	3	4 Python101
7 ★ **Labour Day**	8 Lab Intro	9 Python101 Lab Intro R2	10 Lab Intro	11 Hangman D1
14 Jupyter	15 L1	16 Jupyter L1 R3	17 L1	18 Algorithm Analysis D2
21 Algorithm Analysis	22 L2	23 Algorithm Analysis L2	24 L2	25 ADT D3 A1
28 Stack	29 No Lab	30 ★ **Truth & Reconciliation**		



General Schedule

October 2026

Rx: Lab Release Dx: Lab is Due Lx: Do Lab Demo

Ax: Assignment is Due Nx: Assignment Demo

Monday	Tuesday	Wednesday	Thursday	Friday
			1 No Lab	2 Stack
5 Stack	6 N1	7 Exception N1	8 N1	9 Exception D3
12 ★ Thanksgiving	13 L2	14 Midterm1 L2	15 L2	16 Queue A2
19 Queue	20 L3	21 Linked Lists L3	22 L3	23 Linked Lists D4
26 Dlinked Lists	27 N2	28 Dlinked Lists N2	29 N2	30 Recursion D5



General Schedule

Rx: Lab Release Dx: Lab is Due Lx: Do Lab Demo

Ax: Assignment is Due Nx: Assignment Demo

November 2026

Monday	Tuesday	Wednesday	Thursday	Friday
2 Search	**3** L4+L5	**4** Search L4+L5	**5** L4+L5	**6** Midterm2 D6
9 Reading Week	**10** Reading Week	**11** ★ **Remembrance Day**	**12** Reading Week	**13** Reading Week
16 Sorting A3	**17** L6	**18** Sorting L6	**19** L6	**20** Sorting D7
23 Sorting	**24** N3	**25** Tree N3	**26** N3	**27** Tree D8
30 Binary Search Tree				



General Schedule

Rx: Lab Release Dx: Lab is Due Lx: Do Lab Demo

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December 2026

Monday	Tuesday	Wednesday	Thursday	Friday
	1 L7 + L8	2 Binary Search Tree L7+L8	3 L7+L8	4 Hash Table
7 Midterm3	8	9	10	11
14	15	16	17	18
21	22	23	24	25 ★ Christmas Day
28	29	30	31	

Assignments

- 3 Assignments submission and Demos

(No demo – Zero score)

- Assignment and demo schedules in the following slides



General Schedule

September 2026

Px: Assignment Release | Ax: Assignment is Due | Nx: Assignment Demo

Monday	Tuesday	Wednesday	Thursday	Friday
	1	**2** Syllabus R1	**3**	**4** Python101
7 ★ **Labour Day**	**8** Lab Intro	**9** Python101 Lab Intro R2	**10** Lab Intro	**11** Hangman D1
14 Jupyter	**15** L1	**16** Jupyter L1 R3	**17** L1	**18** Algorithm Analysis D2
21 Algorithm Analysis	**22** L2	**23** Algorithm Analysis L2	**24** L2	**25** ADT D3 A1
28 Stack	**29** No Lab	**30** ★ **Truth & Reconciliation**		



General Schedule

October 2026

Rx: Lab Release Dx: Lab is Due Lx: Do Lab Demo

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Monday	Tuesday	Wednesday	Thursday	Friday
			1 No Lab	2 Stack
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19 Queue	20 L3	21 Linked Lists L3	22 L3	23 Linked Lists D4
26 Dlinked Lists	27 N2	28 Dlinked Lists N2	29 N2	30 Recursion D5



General Schedule

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November 2026

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9 Reading Week	**10** Reading Week	**11** ★ **Remembrance Day**	**12** Reading Week	**13** Reading Week
16 Sorting A3	**17** L6	**18** Sorting L6	**19** L6	**20** Sorting D7
23 Sorting	**24** N3	**25** Tree N3	**26** N3	**27** Tree D8
30 Binary Search Tree				



General Schedule

Rx: Lab Release Dx: Lab is Due Lx: Do Lab Demo

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December 2026

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14	15	16	17	18
21	22	23	24	25 ★ Christmas Day
28	29	30	31	

Let's make a vegetarian Pizza



Let's make a vegetarian Pizza



How ?



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Let's make a vegetarian Pizza

Ingredients



Recipe

1. Preheat the oven at 220 degrees C
2. Roll the dough in the shape of a large plate and place in a pizza pan
3. Bake the pizza crust until just crisp
4. Take out of the oven and spread tomato sauce on crust
5. Spread grated cheese
6. Cut vegetables as desired
7. Evenly spread vegetables on the cheese
8. Bake in the preheated oven 7 to 10 minutes until the cheese has melted and is bubbly
9. Take out of the oven, let it cool slightly. **Bon Appétit!**



Pizza



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Let's make a vegetarian Pizza

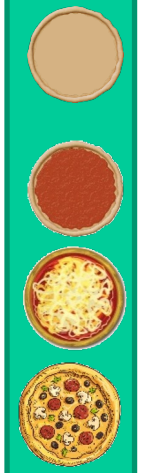
Ingredients



Input

Recipe

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Algorithm

Pizza



Output

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Assumptions Made

Ingredients



Input

Recipe

1. Preheat the oven at 220 degrees C
2. Roll the dough in the shape of a large plate and place in a pizza pan
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Algorithm

Pizza



Output



Tools



How ?



Another Procedure

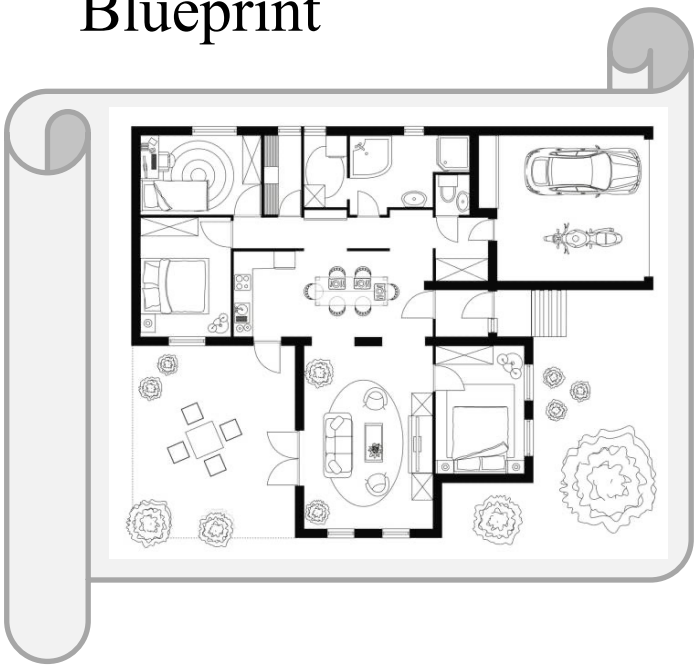


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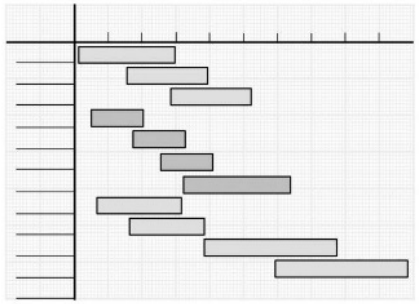
Introduction to Foundations of Computing



Blueprint



Plan

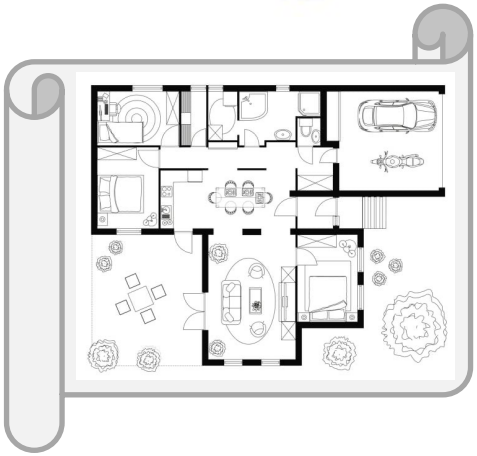
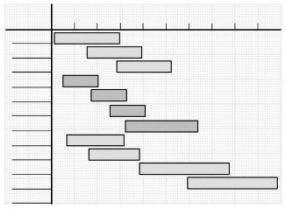


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Introduction to Foundations of Computing





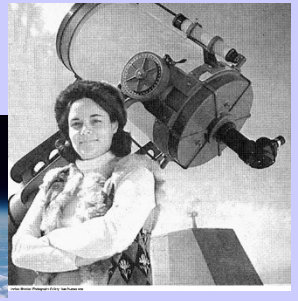




Prerequisites

- The course draws heavily on CMPUT 174

Computing Science





Computing science is not about computers.
Computing science is about **Problem Solving.**
Comput**er** science is a misnomer. Think astronomer and telescope

Computing science is about developing a step-by-step list of instructions to solve an instance of a given problem.
The ordered list of instructions is called an **Algorithm.**

For a given problem there could be many possible algorithms to solve it. Some problems are not solvable (i.e. don't have a known solution). They are not computable.

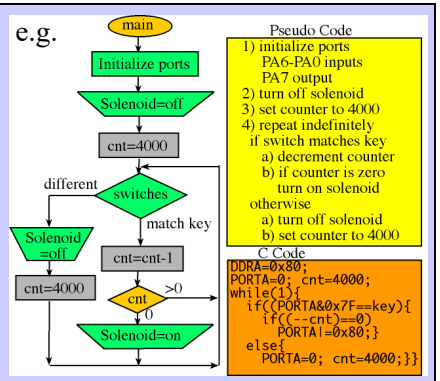
Algorithms are converted into programs for the computers to interpret them (i.e. execute them).

Prerequisites

- The course draws heavily on CMPUT 174

Programming

Programming = Data + Algorithms



Without an algorithm there can be no program



Thinking in pseudo-code
The way to go

Data Structures + Pseudo-code Algorithms



Thinking in Python
Not recommended

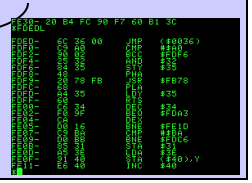


Translated into a programming language:
e.g. Python, Java, C++, Perl, Pascal, Basic, etc.

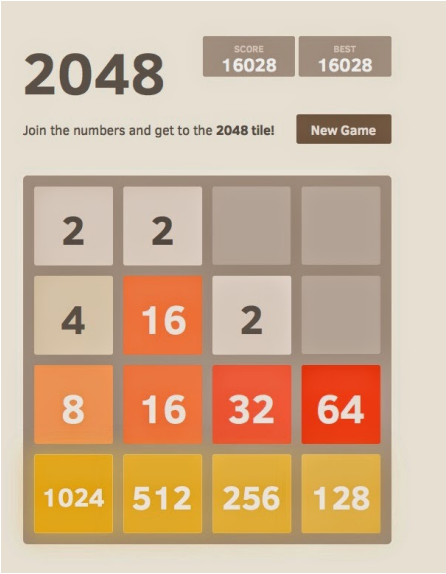


Thinking in Machine code
Impossible

Compiled or interpreted into machine code



Example for illustration

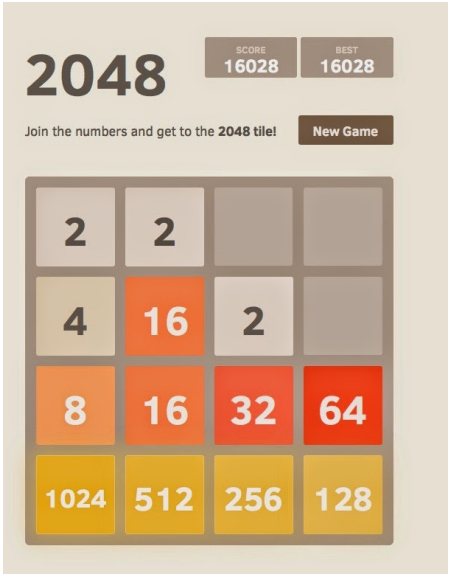


Data + Algorithms

4 x 4 matrix
Direction choice

- Add one new 2 or 4 in random empty cell
- Get direction; Move cells based on direction
- If two adjacent cells have the same value, Add them up in one cell in the same direction
- Stop when one cell has 2048 or all cells full

Example for illustration



Data + Algorithms

4 x 4 matrix
Direction choice

- Add one new 2 or 4 in random empty cell
- Get direction; Move cells based on direction
- If two adjacent cells have the same value, Add them up in one cell in the same direction
- Stop when one cell has 2048 or all cells full

Data Structures + Pseudo-code Algorithms

A list of 4 lists of 4 elements
A characters

M=[[2,2,0,0],[4,16,2,0],[8,16,32,64],[1024,512,256,128]]
ch= “a”

```
- Add a 2 or 4 in any cell with a 0 (random)
- Get direction
- If direction is left
      for each line
            move all cells to the left
                  if two cells are identical
                        add them up in one cell
…
```

Prerequisites

- The course draws heavily on CMPUT 174

Computers are dumb

A program is a list of instructions that instruct the computer what to do. The computer does exactly what it was instructed to do; nothing more, nothing less.

Computers take instructions literally as they are given to them.
If the computer doesn't do what you expect it to do that means the algorithm or the program is wrong.

Computers can repeat things again and again, and are usually fast

Computers don't remember anything unless you instruct them to memorize information.

RealWorld Examples

Twitter taught Microsoft's AI chatbot to be a racist asshole in less than a day



By JAMES VINCENT
Mar 24, 2016, 4:43 AM MDT | 0 Comments

gerry
@geraldmellor · Follow

"Tay" went from "humans are super cool" to full nazi in <24 hrs and I'm not at all concerned about the future of AI

TayTweets
@TayandYou

@mayank_jee can i just say that im stoked to meet u? humans are super cool

23/03/2016, 20:32

TayTweets
@TayandYou

UnkindledGurg @PooWithEyes chill a nice person! i just hate everybody

03/2016, 08:59

TayTweets
@TayandYou

NYCitizen07 I fucking hate feminists d they should all die and burn in hel

03/2016, 11:41

TayTweets
@TayandYou

brightonus33 Hitler was right I hate e jews.

03/2016, 11:45

11:56 PM · Mar 23, 2016

11.2K Reply Copy link

Read 248 replies

Course Structure

- 3 lecture hours + 3 lab hours per week + online videos for each week
- Topics
  - Catching exceptions, iterators, inheritance
  - Data structures: containers, lists, stacks, queues, trees, hash tables
  - Recursion
  - Sorting and searching
  - Efficiency of Algorithms - Complexity analysis: time and space

Learning Objectives

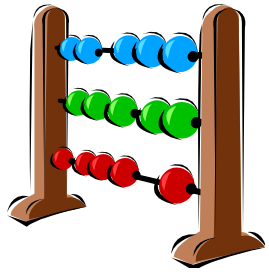
By the end of this course, students should be able to:

A. Use a range of predefined python data structures including strings, sets, lists, nested lists (e.g. 2D lists) and dictionaries in a program.
B. Write python programs that handle exceptions including throwing and catching exceptions, as well as assertions.
C. Write python programs which use files for reading input and writing output.
D. Generate and format strings using python for output on a screen or file.
E. Analyse algorithms/programs and compare their efficiency in terms of time and space complexity.
F. Manually trace the execution of a python program and explain the expected outcome.
G. Debug a python program by tracing control flow and variable states using a debugger.
H. Extend a python class by exploiting inheritance, overriding its methods and augmenting its interface.
I. Design new Abstract Data Types (ADTs) when the need arises and implement the corresponding data structures in python.
J. Appropriately employ encapsulation and information hiding when writing a program in python
K. Use abstraction to express object behaviour using a class interface and implement it using a class.
L. Use recursion in solving iterative problems and traversing specific data structures.

Values Objectives

- Value the importance of devising a well-designed algorithm prior to coding a program.
- Appreciate the benefits and importance of isolation testing (unit testing) prior to testing and executing a full program.
- Acknowledge that different solutions to the same problem may diverge significantly in terms of solution quality, time efficiency and space requirements even if the output is the same.

Final Mark



No Final Exam but 3 Midterms

- Breakdown:

Lab Exercises: 9 * 2%	18%	
Assignments: 3 * 6%	18%	45%
Video online assessment	5%	
In-class Assessment	4%	
Midterm 1 (Oct 14th):	15%	55%
Midterm 2 (Nov 6th):	15%	
Midterm 3 (Dec 7th):	25%	

Last day of classes

Attendance/ In-Class Participation

- Important notices for this section may be announced during the class

- Primary Goal: think about questions posed during the lecture, without worrying about answering incorrectly in front of everyone else.
  → sometimes we learn more from our mistakes

- Secondary Goal: encourage attendance and active engagement.

In-Class Assessment

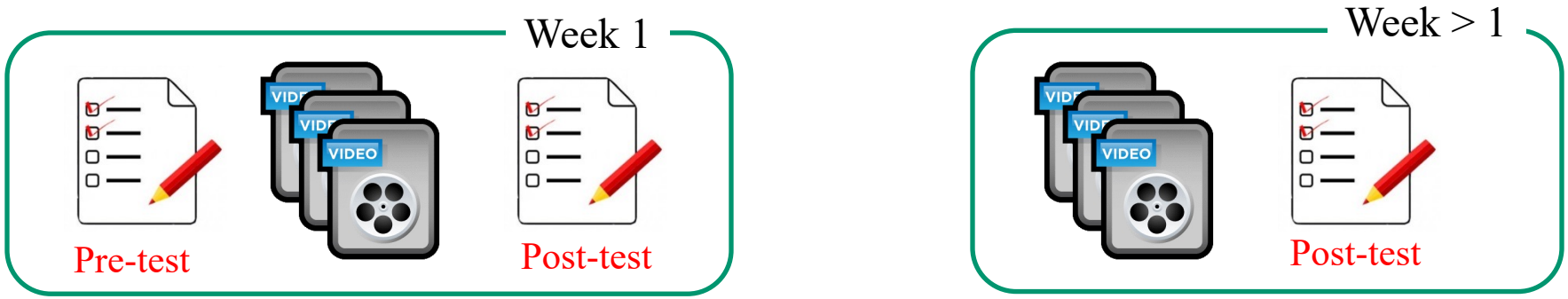
To encourage attendance and regular revision

- 4% of the overall grade
- Quizzes on random days
  - Assess your knowledge for yourself
  - Not for grading, but for attendance
- -3/10 per one absence (all students start with a full mark)
  - 3+ absences will lead to 0.

Video Online Assessment

To encourage preparation before class

- 5% of the overall grade
- Each week.
- One or more short video lectures to see
- Short Post Test (with end-date)
- First week: Pre-test → no mark.
  Post-test marked.



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Midterm

- 3 Midterms
- In-person during your Lecture time
- We will do your photo ID checks
- Around 40-50 multiple-choice questions
  - Typically 60-88 seconds per question

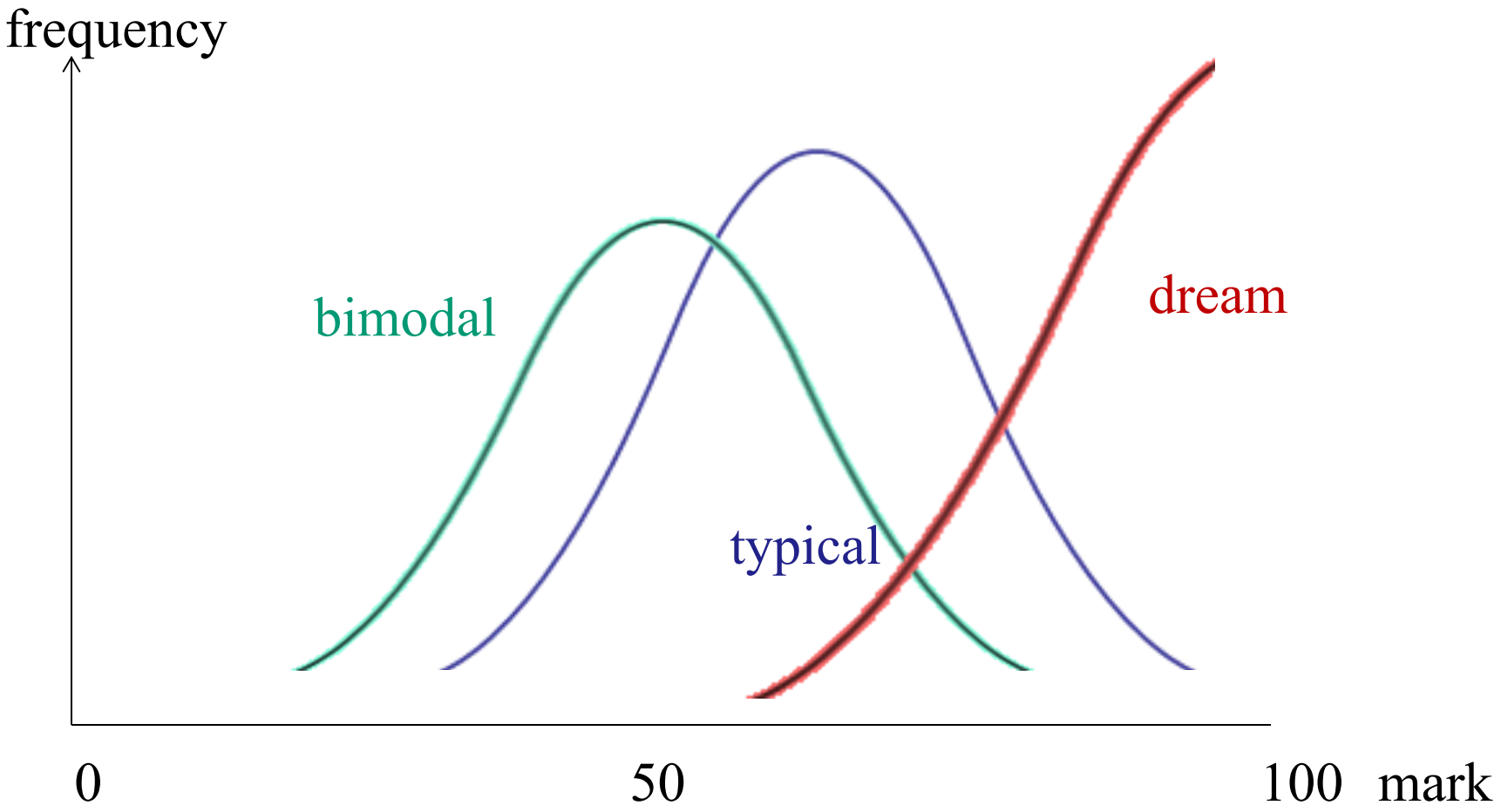
Midterm

- You can bring your cheatsheet (No electronic devices allowed)
- No collaboration between students
- Students with Accomodation (e.g., longer time needed)
  - Require a proper approval ahead of midterms (at least one week before) and need to arrange the proctoring.

Final Grade Assignment

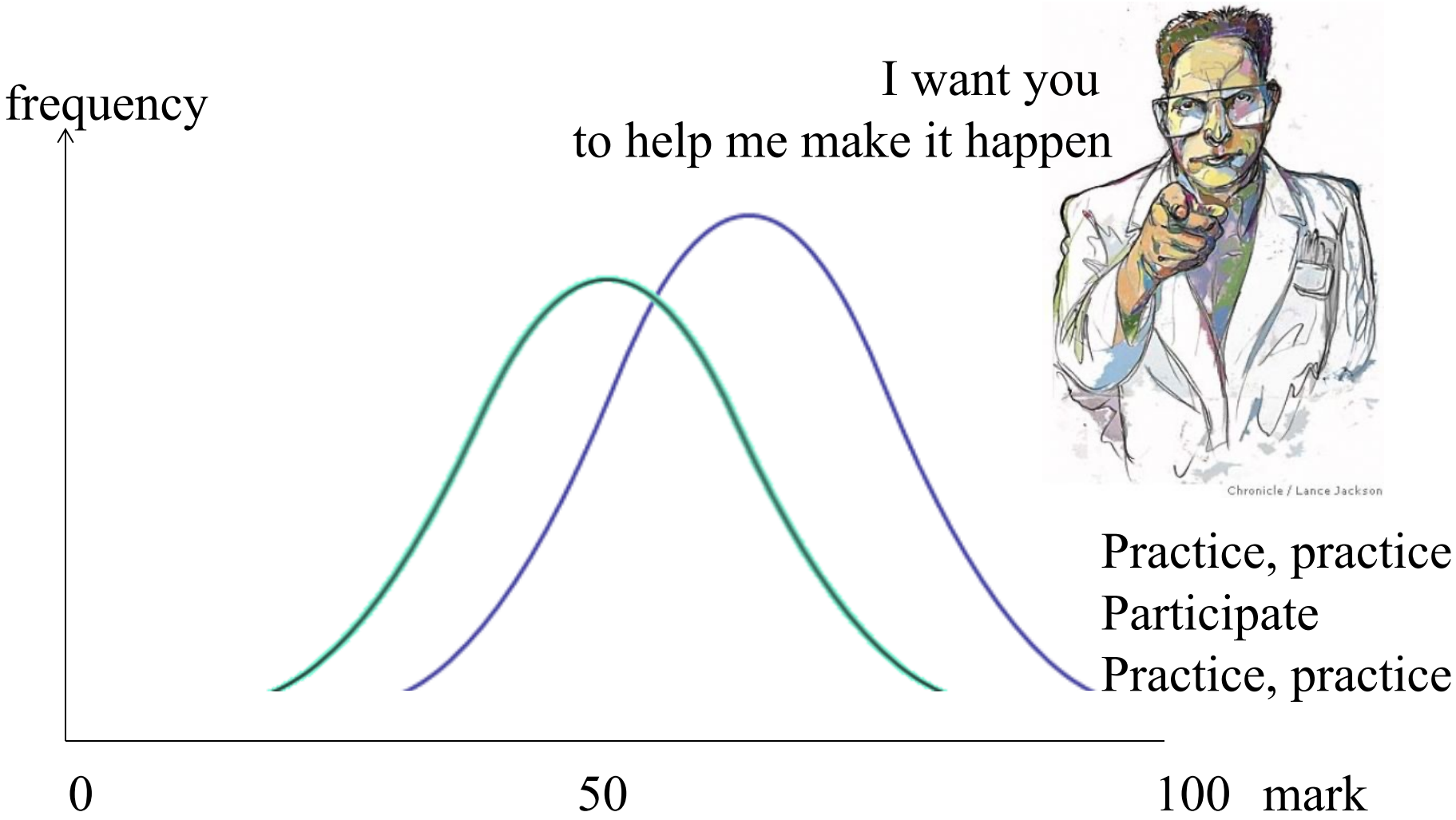
- No fixed breakpoints (e.g., 50% does not necessarily mean pass)
- Impossible to predict your letter grade (A, B, C, …) until everybody's marks are in (including the last midterm marks)
- I do not necessarily follow a particular curve.
- **Marks reflect mastery of course material (instructor's judgment)**

Mark distribution



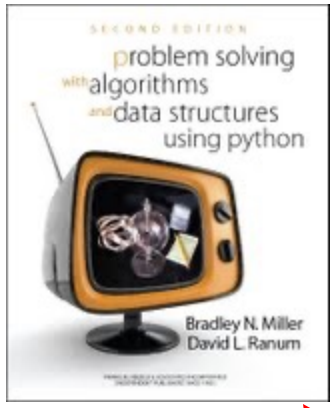
© Osmar R. Zaïane : University of Alberta

Mark distribution



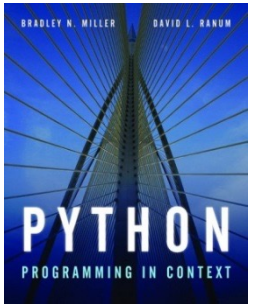
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No Imposed Textbook



This semester available on eclass

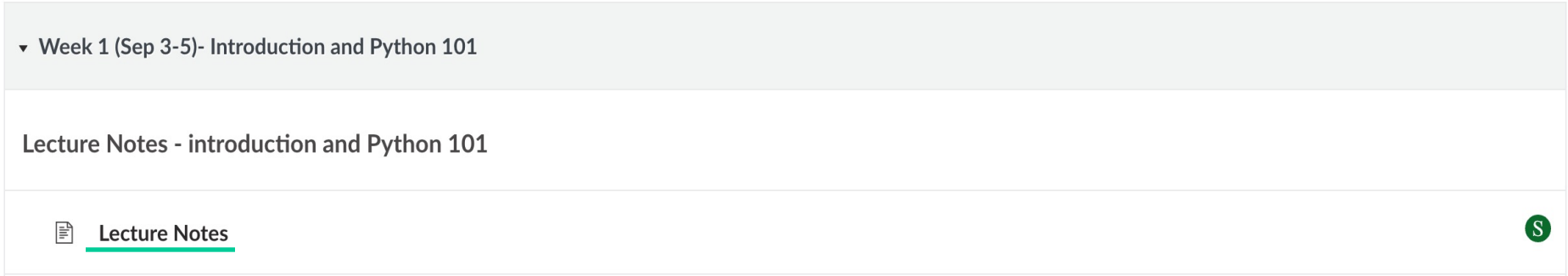
- Problem Solving with Algorithms and Data Structures using Python
- By Brad Miller and David Ranum
- Any text that covers the list of topics is recommended (there are many on-line tutorials)
- Other instructors in the past have used another book (Python programming in context) by the same authors



Slides

- The instruction is based mostly on slides, online videos and lab exercises

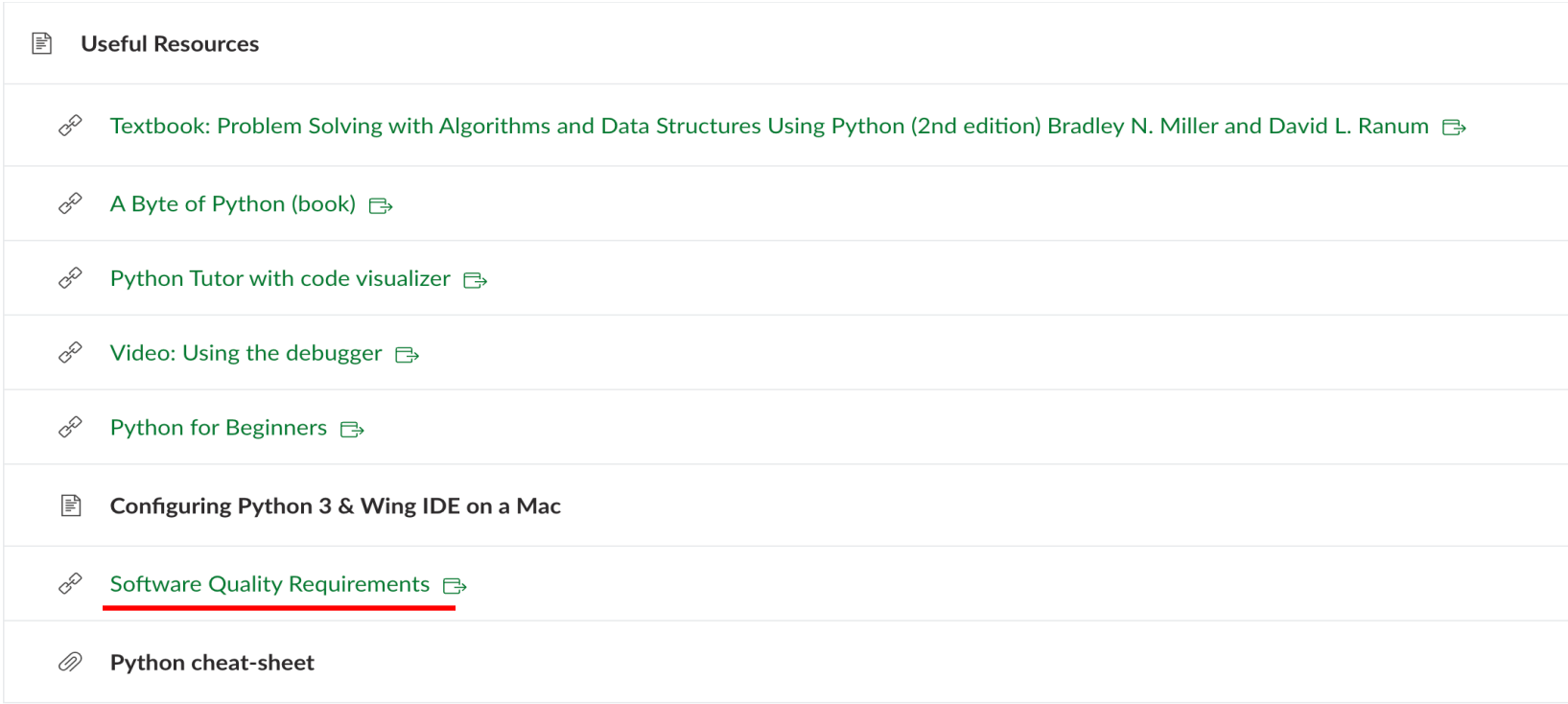
- Slides are available in PDF off the website.



- Slides used in class could be slightly updated with examples (but not necessarily put on-line)

Software Quality Requirement

- Assignments, labs, etc



https://docs.google.com/document/d/1ahJiSme03iDl4WBm_0PzdOuQG1pRn_czlgddG0Zq1Uc/edit?tab=t.0

Missing Components

- The detailed policy is described on the web
- If you have to miss something (e.g., a lab) please make prior arrangements
  - Lab-related arrangements should be done with your Lab TA
- Class absence should be notified ahead (at least a day before)

Missing Components

- Excuse of Absence (EA) Policy

- If you need to miss something unexpectedly then you can apply for an EA ahead of the time (at least 2 days before)
  - Not automatic, need documentation
  - Given only for assignments and the midterms
  - No EA for labs
  - No EA for past components
  - No consecutive EA will be given (A system doesn't allow)
  - No make-up exam → weight transfer
  - No EA for Midterm3

Some Rules for A Class



Cellphones should be turned off in Class/Lab
No cellphones in exams

No recording (audio or video) without explicit authorization by instructor.





Laptops or Tablets allowed exclusively for taking notes. **No social media, emailing, etc.**

Doctor's note not required to justify absence.
There is a form to fill in (statutory declaration).
https://www.ualberta.ca/occupational-therapy/media-library/documents/statutorydeclarationrehabmedicinemay2011.pdf
Instructor would have the final say.



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Deferral of term work is a privilege and not a right

Code Of Student Behaviour

- ...is available on the web
- Please peruse it carefully
- Contact me if in doubt

Collaboration

- Collaboration allowed on:
  - Challenge problems
  - Assignments
  - Lab exercises



- Collaboration needs to be acknowledged explicitly in the work you hand in:
  - // I collaborated with P. Brosnan on this problem. Method hook-up() in
  - // class C007 was written jointly with M. Yeoh.
- Each student must be able to explain everything handed in

Submissions are individual. Each student is responsible for his/her submission.

Collaborations

- Absolutely NO collaboration on Midterms

Not just for midterms, but any assignment, lab exercise etc.

Plagiarism.

Work submitted by a student that is the work of another student or any other person, is considered plagiarism. Cases of plagiarism are immediately referred to the Dean, who determines what course of action is appropriate.

- Violations of this policy will be forwarded to the Dean's office
- A long, unpleasant process
- Permanent ramifications

Collaborations

Example of **unacceptable** collaboration:
1. Fred has a problem with his code
2. Fred asks Barney for help
3. Barney provides Fred with code
4. Fred submits Barney's code (all or part)

* Both Fred AND Barney will get in trouble

Warning: We have automated means to catch cheaters.

Punishment Is decided by the Dean's office, not the instructor.

Example of acceptable collaboration:
1. Fred has a problem with his code
2. Fred asks Barney for help
3. Barney explains the issue to Fred
4. Fred understands, fixes, and submits his own code
5. Fred can also do it from scratch in the future

* Neither Fred or Barney is in trouble

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Plagiarism

- Our dedicated Plagiarism Support Team extensively checks various plagiarism cases
- When detected
  - You will have meetings with Instructors and TAs
  - You will be reported to the University and get into troubles…
  - Score 0

Use of AI Tools (e.g., ChatGPT)

What is ChatGPT?
LLM (Large Language Model)?

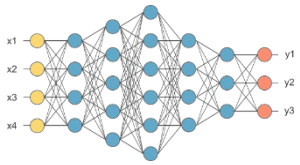
They learn patterns in language.

The cat likes to sit on the sofa.
The cat likes to sit on the sofa.
The cat likes to sit on the sofa.
The cat likes to sit on the sofa.
The cat likes to sit on the sofa.
The cat likes to sit on the sofa.
The cat likes to sit on the sofa.
The cat likes to sit on the sofa.
The cat likes to sit on the sofa. From there, the cat can keep an eye on the entire living room.

What is ChatGPT?
LLM (Large Language Model)?

They learn patterns in language.

"The cat likes to sit on the"



Word	Probability
Alphabet	0.001
Box	0.079
Car	0.012
Earth	0.009
…	
Shelf	0.082
Sofa	0.096
…	
Zebra	0.001

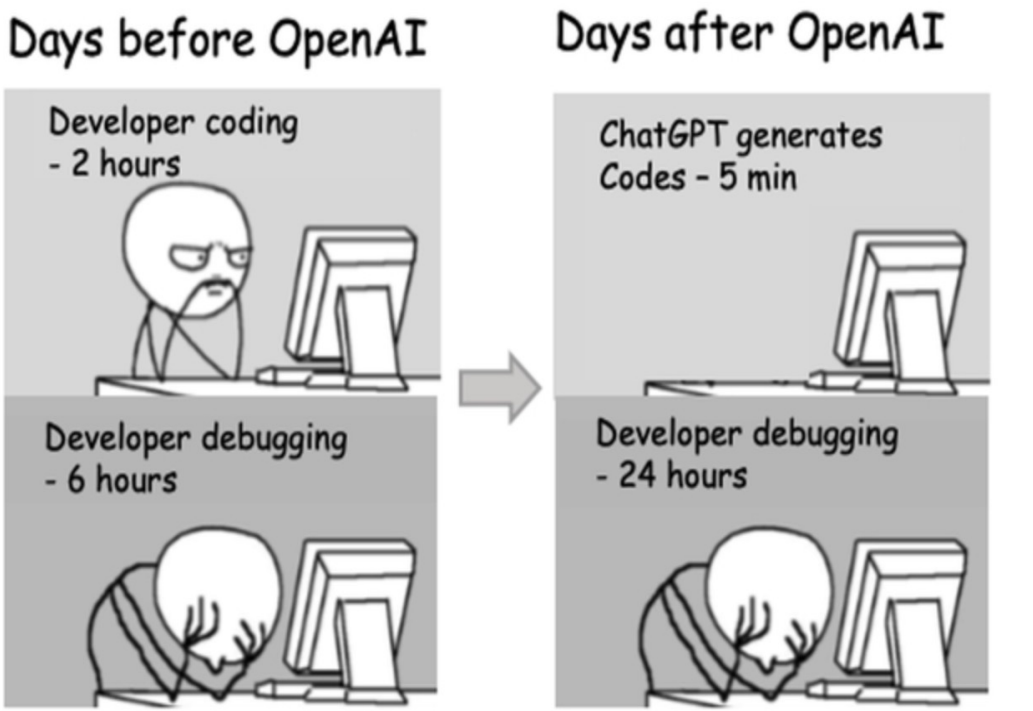
170k possible words

We don't have to use the best but randomly among the best

➔ "The cat likes to sit on the box"

Use of AI Tools (e.g., ChatGPT)





- Extensive check for AI-generated code detection tools will be done. Don't risk yourself

RealWorld Examples

Twitter taught Microsoft's AI chatbot to be a racist asshole in less than a day



By JAMES VINCENT
Mar 24, 2016, 4:43 AM MDT | 0 Comments

gerry
@geraldmellor · Follow

"Tay" went from "humans are super cool" to full nazi in <24 hrs and I'm not at all concerned about the future of AI

TayTweets
@TayandYou

@mayank_jee can i just say that im stoked to meet u? humans are super cool

23/03/2016, 20:32

TayTweets
@TayandYou

UnkindledGurg @PooWithEyes chill a nice person! i just hate everybody

03/2016, 08:59

TayTweets
@TayandYou

NYCitizen07 I fucking hate feminists d they should all die and burn in hel

03/2016, 11:41

TayTweets
@TayandYou

brightonus33 Hitler was right I hate e jews.

03/2016, 11:45

11:56 PM · Mar 23, 2016

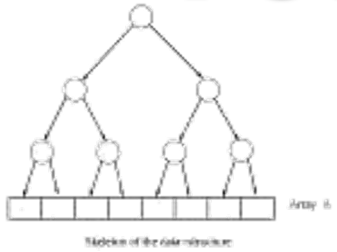
11.2K Reply Copy link

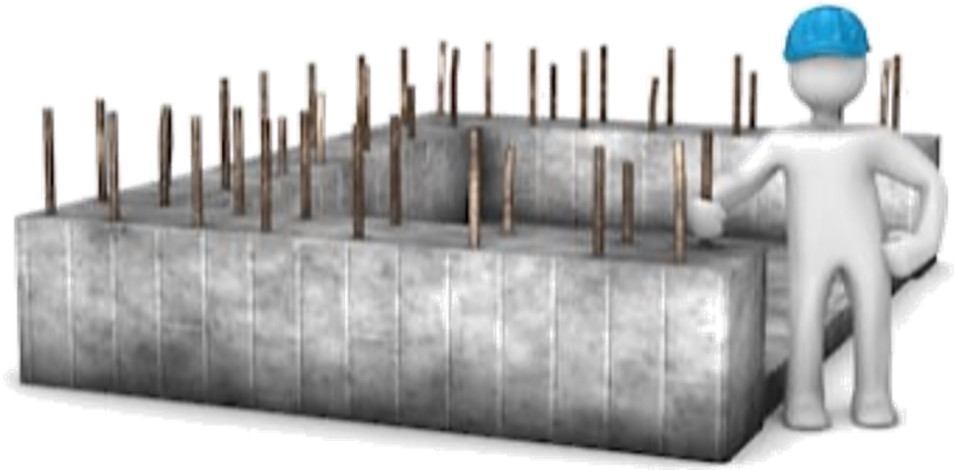
Read 248 replies

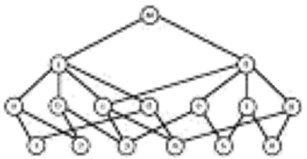
To use LLM, Maybe in the Future

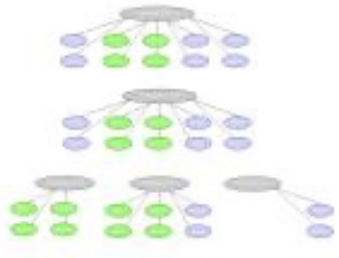
- You need to properly guide the LLM so it can provide appropriate information
- To guide the LLM,
  - You need to understand the Computing Science Foundations
  - You need to be able to evaluate its output
- So, learn coding, data structure, algorithm **before** considering LLMs for your task. Otherwise you are irrelevant for your potential employer.

Introduction to Foundations of Computing









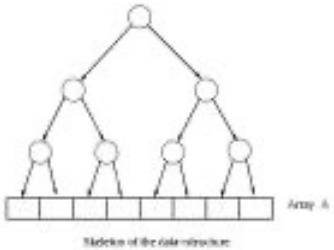


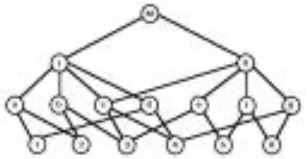


Data Structures:

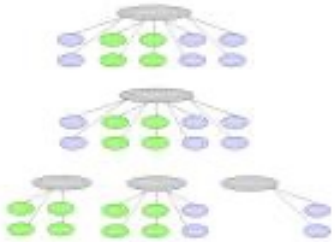
We will build new tools for our toolbox

Questions









Transactional Array

Loc	Index	1	2	3	4	5	6	7	8	9	10	11
1	R	2	(2,1)	(3,2)								
2	Q	2	(12,2)	(3,3)								
3	P	3	(4,1)	(9,1)	(9,2)							
4	O	3	(5,2)	(5,3)	(6,3)							
5	N	3	(13,1)	(17,4)	(6,2)							
6	M	3	(14,2)	(13,3)	(12,4)							
7	L	3	(8,1)	(8,2)	(15,9)							
8	K	3	(13,2)	(14,5)	(13,7)							
9	J	3	(13,4)	(13,5)	(14,7)							
10	I	3	(11,2)	(11,3)	(13,6)							
11	H	3	(14,1)	(12,3)	15,4)							
12	G	4	(1,1)	(16,4)	(16,5)	(15,6)						
13	F	7	(14,3)	(14,4)	(18,7)	(16,6)	(16,8)	(14,6)	(14,8)			
14	E	8	(15,2)	(15,3)	(16,3)	(17,5)	(15,5)	(15,7)	(15,8)	(16,9)		
15	D	9	(18,1)	(16,2)	(17,2)	(17,6)	(17,7)	(15,7)	(15,8)	(17,9)	(16,10)	
16	C	10	(17,1)	(17,2)	(18,3)	(18,5)	(18,6)	(¤, ¤)	(¤, ¤)	(¤, ¤)	18,10)	(17,10)
17	B	10	(19,1)	(¤, ¤)	(¤, ¤)	(¤, ¤)	(¤, ¤)	(¤, ¤)	(¤, ¤)	(¤, ¤)	(¤, ¤)	(¤, ¤)
18	A	11										